

# Business Intelligence Overview Report

## Superstore Sales Performance Dashboard

*Business Understanding & Data Preparation*

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Dataset

**Superstore Sales Dataset**

Project Period

**Week 2 – AnalystLab Africa Data Analytics Internship Program**

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# Part 1 – Business Intelligence Overview Report

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## 1. What is Business Intelligence?

Business Intelligence (BI) is the process of collecting, analysing, and presenting business data in a way that helps organisations make better decisions. It turns raw data into useful information through reports, dashboards, and visualisations.

In simple terms, BI answers important business questions such as:

- How are we performing?
- Where are we making or losing money?
- Which customers, products, or regions are driving results?
- What actions should we take next?

BI combines data from different sources, organises it, and delivers clear insights to managers and executives.

## 2. Why Organisations Use Dashboards

Dashboards are one of the most important tools in Business Intelligence. They display key performance indicators (KPIs) and trends on a single screen in a visual and easy-to-understand format.

Organisations use dashboards because they:

- **Provide a quick overview of business performance** without reading long reports.
- **Highlight important metrics** such as sales, profit, orders, and margins.
- **Allow interactive analysis** through filters and slicers (e.g. by region, segment, or time period).
- **Support faster decision-making** by showing real-time or up-to-date information.
- **Improve communication** between teams and management by presenting a single version of the truth.

For executives, a well-designed dashboard acts like a control panel for the business.

## 3. How Dashboards Improve Decision-Making

Dashboards improve decision-making in several practical ways:

1. **Speed** – Managers can see performance immediately instead of waiting for manual reports.
2. **Clarity** – Charts and KPI cards make complex data easier to understand.
3. **Focus** – Key metrics are prioritized so attention is directed to what matters most.

4. **Exploration** – Interactive filters allow users to dig deeper (for example, comparing regions or product categories).
5. **Evidence-based decisions** – Recommendations are based on actual data rather than assumptions.

When decision-makers can clearly see what is working and what is not, they can respond faster and more confidently.

#### 4. Business Objective of the Selected Dataset

The selected dataset is the **Superstore Sales Dataset**, which contains sales transactions for a national retail company operating across the United States between 2014 and 2017.

The main business objective of this project is:

*To design an interactive executive dashboard that enables management to monitor sales performance, profitability, customer behaviour, and regional performance, and to provide data-driven recommendations for improving business results.*

Specifically, the dashboard will help answer questions such as:

- What are the overall sales and profit performance?
- Which regions and customer segments generate the most revenue?
- Which product categories and products are most profitable?
- What trends can be observed over time?
- Where should management focus to improve performance?

By transforming the raw sales data into a clear and interactive dashboard, the company will be better positioned to make informed strategic decisions.

## Part 2 – Data Preparation

### 1. Data Import

The Superstore Sales dataset was imported into Microsoft Power BI using the Excel connector. The file contains **9,994 rows** and **21 columns**.

### 2. Dataset Structure Review

| Column        | Data Type      | Description                                         |
|---------------|----------------|-----------------------------------------------------|
| Row ID        | Whole Number   | Unique row identifier                               |
| Order ID      | Text           | Unique order number                                 |
| Order Date    | Date           | Date the order was placed                           |
| Ship Date     | Date           | Date the order was shipped                          |
| Ship Mode     | Text           | Shipping method                                     |
| Customer ID   | Text           | Unique customer identifier                          |
| Customer Name | Text           | Customer full name                                  |
| Segment       | Text           | Customer segment (Consumer, Corporate, Home Office) |
| Country       | Text           | Country (United States)                             |
| City          | Text           | City name                                           |
| State         | Text           | State name                                          |
| Postal Code   | Whole Number   | Postal code                                         |
| Region        | Text           | Region (West, East, Central, South)                 |
| Product ID    | Text           | Unique product identifier                           |
| Category      | Text           | Product category                                    |
| Sub-Category  | Text           | Product sub-category                                |
| Product Name  | Text           | Full product name                                   |
| Sales         | Decimal Number | Sales amount                                        |
| Quantity      | Whole Number   | Quantity sold                                       |
| Discount      | Decimal Number | Discount rate                                       |
| Profit        | Decimal Number | Profit amount                                       |

### 3. Missing Values Check

A thorough check was performed in Power Query. Result: there are **no missing values** in any column.

### 4. Duplicate Records

- Fully duplicate rows: 0

- Duplicate Order IDs exist, but this is expected and correct (one order can contain multiple products).

**Action taken:** No rows were removed.

## 5. Data Type Corrections

Most data types were already appropriate after import. The following checks and minor adjustments were made in Power Query:

| Column      | Action                                             |
|-------------|----------------------------------------------------|
| Order Date  | Confirmed as Date                                  |
| Ship Date   | Confirmed as Date                                  |
| Sales       | Confirmed as Decimal Number                        |
| Profit      | Confirmed as Decimal Number                        |
| Discount    | Confirmed as Decimal Number                        |
| Quantity    | Confirmed as Whole Number                          |
| Postal Code | Kept as Whole Number (acceptable for this dataset) |

## 6. Basic Cleaning & Transformations (Power Query)

The following light transformations were applied:

1. **Removed unnecessary column** – Row ID was removed because it has no analytical value.
2. **Added helpful columns for analysis** – Year and Month (extracted from Order Date), Month Name (for clearer charts), and Profit Margin ( $\text{Profit} \div \text{Sales}$ ).
3. **Renamed columns** slightly for clarity where needed.
4. **Checked for leading/trailing spaces** in text columns – none found.

## 7. Final Dataset Summary (After Preparation)

| Item           | Value                                                  |
|----------------|--------------------------------------------------------|
| Total Rows     | 9,994                                                  |
| Total Columns  | 20 (after removing Row ID) + 4 new helper columns = 24 |
| Missing Values | 0                                                      |
| Duplicate Rows | 0                                                      |
| Date Range     | 3 Jan 2014 – 30 Dec 2017                               |

The dataset is now clean, properly typed, and ready for dashboard development in Power BI.